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MINOR PROJECT 2
(15B19CI591) PROJECT SYNOPSIS

TITLE OF PROJECT:

"Graph Attention Networks for Land Cover Classification from Multispectral Satellite Images"

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ABSTRACT

Land cover classification using satellite imagery plays an important role in **remote sensing and environmental monitoring**. It helps identify different surface types such as forests, water bodies, urban regions, and agricultural land. Accurate land cover information is useful for understanding environmental changes, monitoring urban expansion, managing agricultural activities, and studying the effects of climate change.

Traditional machine learning techniques for land cover classification mainly rely on **pixel-level spectral features** extracted from satellite images. While these methods can identify basic patterns, they often struggle to capture the **spatial relationships between neighboring regions** within an image. As a result, their performance may be limited when different land cover types have similar spectral characteristics.

Deep learning approaches, particularly **Convolutional Neural Networks (CNNs)**, have improved classification performance by learning more complex features from satellite imagery. However, CNN-based models primarily focus on local patterns in the image and may not fully capture the relationships between different regions across the entire scene.

To address this limitation, this project explores the use of **Graph Neural Networks (GNNs)** for land cover classification. In this approach, satellite images are represented as a **graph structure**, where different regions of the image act as nodes and their spatial connections represent relationships between them. This representation helps the model better understand how neighboring areas influence each other.

For each region in the image, important information such as **spectral values from multiple satellite bands** and vegetation indicators like the **Normalized Difference Vegetation Index (NDVI)** is extracted. The Graph Neural Network then processes this information while considering the spatial relationships between regions, allowing it to learn more meaningful patterns in the data.

Using this approach, the system can classify different land cover categories such as **vegetation, water bodies, urban areas, and agricultural land**. The proposed method can support a variety of practical applications, including **environmental monitoring, urban planning, agricultural assessment, and disaster management**, by providing more reliable land cover classification from satellite imagery.

INTRODUCTION

Remote sensing using satellite imagery has become an essential tool for monitoring Earth's surface. Land cover classification helps identify different types of surfaces such as forests, water bodies, urban areas, and agricultural land. Accurate land cover mapping plays an important role in environmental monitoring, agricultural planning, climate studies, and urban development.

Traditional methods for land cover classification include techniques such as:

- Support Vector Machines
- Random Forest classifiers
- Pixel-based spectral classification

These methods primarily rely on **spectral features** extracted from satellite images. While they can provide reasonable classification accuracy, they often fail to effectively capture **spatial relationships between neighboring regions** in an image. As a result, they may struggle to distinguish between land cover types that have similar spectral characteristics but different spatial patterns.

To address these limitations, **deep learning techniques** such as Convolutional Neural Networks (CNNs) have been widely used. CNNs improve classification performance by automatically learning hierarchical features from satellite images. They are effective at detecting local patterns such as textures and edges. However, CNNs mainly capture **local spatial patterns** and may not effectively represent complex relationships between distant regions in large satellite images.

Graph Neural Networks (GNNs) provide a powerful alternative for modeling such relationships. In GNN-based approaches, image data is represented as a **graph structure**, where spatial relationships between regions can be explicitly modeled. In this representation:

- Image regions or superpixels become **nodes**.
- Spatial relationships between neighboring regions become **edges**.

By representing satellite imagery as a graph, GNNs can capture both **spectral features and spatial dependencies** between different regions. This allows the model to learn more meaningful representations of land cover patterns.

In this project, we apply **Graph Neural Networks (GNNs)** to multispectral satellite images. The proposed system integrates spectral information from satellite bands, vegetation indices such as NDVI, and spatial relationships between image regions. By combining these different types of information, the model can better understand the structure of the landscape and improve classification accuracy.

GRAPH NEURAL NETWORKS

Graph Neural Networks (GNNs) are a class of deep learning models designed to process data that can be represented as graphs. In a graph, data points are represented as **nodes**, while the relationships between them are represented as **edges**. GNNs learn by iteratively aggregating information from neighboring nodes, allowing each node to update its representation based on both its own features and the features of its neighbors.

One of the key strengths of GNNs is their ability to model **complex spatial relationships** in structured data. In satellite imagery, different land cover regions often influence each other. For example, agricultural fields may appear near irrigation channels, forests may border rivers, and urban areas may be connected to road networks. By capturing these relationships, GNNs can better understand the context of each region.

Another important advantage of GNNs is **efficient feature aggregation**. During training, each node gathers information from its neighboring nodes and combines it with its own features. This process allows the network to gradually build richer and more informative representations of each region in the image. As multiple layers of the network are applied, the model can capture information from increasingly larger neighborhoods within the graph.

GNNs also support the **integration of multiple feature types**. In remote sensing applications, nodes may contain spectral information from different satellite bands, vegetation indices such as NDVI, or texture-based features. By combining these features with spatial relationships between nodes, GNNs can create a more comprehensive representation of the landscape.

Another key benefit is **improved classification performance**. By considering both spectral characteristics and spatial context, GNN-based models can more accurately distinguish between land cover classes that might appear similar in traditional pixel-based approaches. This leads to improved performance in tasks such as land cover classification, vegetation monitoring, and environmental analysis.

In summary, Graph Neural Networks provide a powerful framework for analyzing satellite imagery by representing image regions as graph nodes and modeling their spatial relationships. This capability enables more effective integration of spectral and spatial information, leading to improved accuracy and reliability in land cover classification systems.

FORMULAS

1. Node Feature Aggregation in Graph Neural Networks

In **Graph Neural Networks (GNNs)**, each node updates its representation by aggregating information from its neighboring nodes. This process allows the model to capture spatial relationships between connected regions in the graph.

$$h_i' = \sigma\left(\sum_{j \in N_i} W\, h_j\right)$$

This equation represents the **feature update process** for node iii. The features from neighboring nodes are combined and transformed to generate an updated representation.

Where:

- h_i' = updated feature representation of node i
- h_j = feature vector of neighboring node j
- W = learnable weight matrix applied to node features
- N_i = set of neighboring nodes of node i
- σ = activation function

Through multiple layers of this aggregation process, nodes gradually incorporate information from a larger neighborhood within the graph.

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2. Node Representation Update

After aggregating information from neighboring nodes, the node's own features are combined with the aggregated neighbor features to create a more informative representation.

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \mathrm{AGG}\left(\{h_j^{(l)} : j \in N_i\}\right)\right)$$

This equation represents the **layer-wise update rule** used in many Graph Neural Networks.

Where:

- $h_i^{(l)}$ = feature vector of node i at layer l
- $h_i^{(l+1)}$ = updated feature vector at the next layer
- $AGG(\cdot)$ = aggregation function (such as mean, sum, or max)
- $W^{(l)}$ = learnable weight matrix at layer l
- σ = nonlinear activation function

This iterative updating process allows nodes to gradually capture both **local and global structural information** from the graph.

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3. NDVI Vegetation Index

Vegetation detection in satellite imagery is commonly performed using the **Normalized Difference Vegetation Index (NDVI)**. It measures vegetation health by comparing the reflectance of near-infrared and red light.

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Where:

- NIR = reflectance value in the Near-Infrared band
- Red = reflectance value in the Red spectral band

Healthy vegetation reflects more near-infrared light and absorbs more red light, resulting in **higher NDVI values**. This index helps distinguish vegetation from other land cover types such as water, soil, or urban surfaces.

METHODOLOGY

1. Problem Statement

The objective of this project is to develop a system that classifies land cover types from multispectral satellite images using Graph Attention Networks while capturing spatial relationships between image regions.

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2. Dataset Preparation

The project uses the **EuroSAT multispectral satellite dataset**.
The dataset contains **10 land cover classes**, including:

• Forest
• Residential
• Industrial
• River
• Sea/Lake
• Highway
• Pasture
• Annual Crop
• Permanent Crop
• Herbaceous Vegetation

Each image contains **13 spectral bands captured by Sentinel-2 satellites**.

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3. Data Preprocessing

The preprocessing pipeline includes:

- **Image Loading**

  Satellite images are loaded from the dataset directory.
- **Image Resizing**

  All images are resized to 64 × 64 resolution.
- **Normalization**

  Pixel values are normalized to improve training stability.
- **RGB Visualization**

  RGB images are generated from multispectral bands for visualization.
- **NDVI Calculation**

  Vegetation indices are computed using the NIR and Red bands.

- **Superpixel Segmentation**

  Images are segmented into superpixels using the SLIC algorithm.
- **Feature Extraction**
  Spectral and vegetation features are extracted for each superpixel.

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4. Graph Construction

After segmentation:
• Each superpixel becomes a **graph node**
• Adjacent superpixels form **graph edges**
• Node features represent spectral and vegetation information

This graph representation allows the Graph Attention Network to learn spatial relationships between regions.

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5. Model Training

The Graph Attention Network is trained using the **extracted node features**.

Training includes:
• Node feature initialization
• Attention coefficient computation
• Neighbor feature aggregation
• Node classification

Optimization details:
Loss Function: Cross-Entropy Loss
Optimizer: Adam optimizer

Evaluation metrics include:
• Classification Accuracy
• F1-Score
• Intersection over Union (IoU)

FLOWCHART

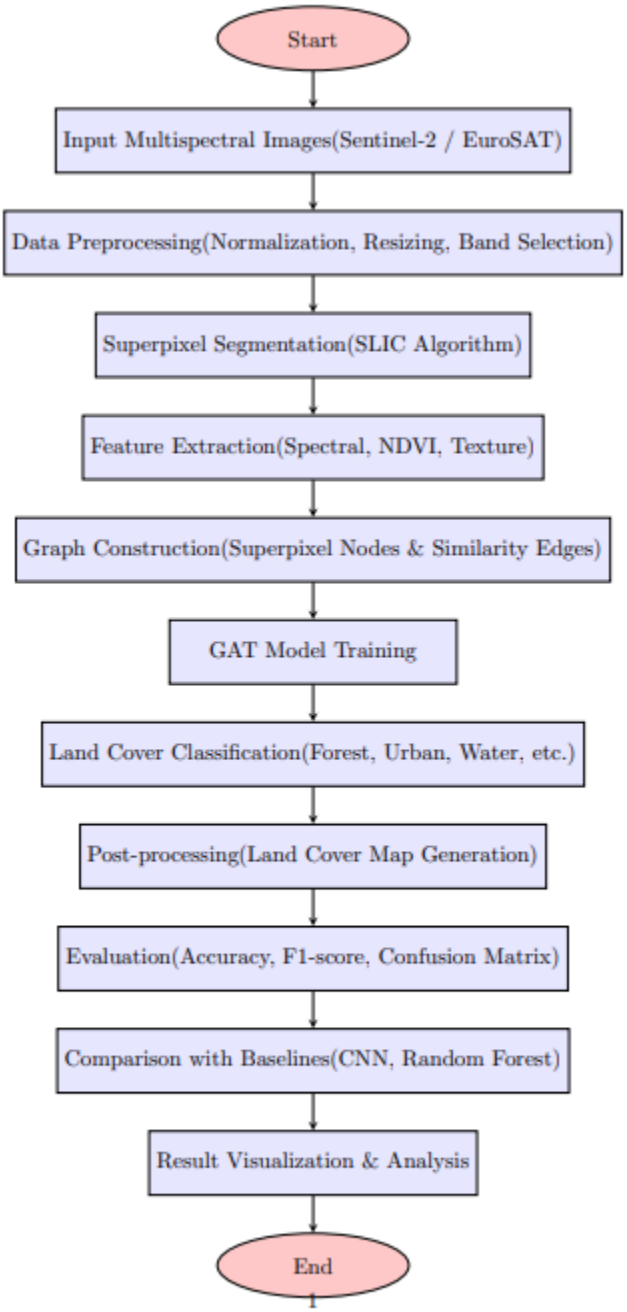

Figure 1: Methodology Flowchart for GAT Land Cover Classification

KEY FEATURES

1. Multispectral Satellite Image Processing

Multispectral satellite imaging involves capturing Earth's surface data across **multiple wavelength bands**, including visible light (red, green, blue) and non-visible bands such as **near-infrared** (**NIR**) and **shortwave infrared** (**SWIR**). These additional spectral bands allow the detection of features that cannot be observed through normal photography.
For example, healthy vegetation reflects more near-infrared radiation compared to stressed or unhealthy vegetation. By analyzing these spectral bands, researchers can monitor vegetation health, soil moisture, land cover, and environmental changes. Multispectral processing forms the foundation for advanced remote sensing analysis, enabling accurate classification of land surfaces and environmental monitoring.

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2. NDVI-Based Vegetation Analysis

The **Normalized Difference Vegetation Index (NDVI)** is one of the most widely used indicators for evaluating vegetation health using satellite imagery. NDVI works by comparing the reflectance values of the **near-infrared (NIR)** band and the red band.
NDVI = {NIR - Red}/{NIR + Red}

Healthy plants absorb most visible red light for photosynthesis and reflect a large portion of near-infrared light. This difference allows NDVI to measure vegetation density and condition.

Typical NDVI values range between **-1** and **+1**:

- 0.6 to 1.0 – Dense and healthy vegetation
- 0.2 to 0.5 – Moderate vegetation
- 0 to 0.2 – Bare soil or sparse vegetation
- Below 0 – Water, snow, or clouds

NDVI is commonly used in **agriculture monitoring, drought detection, forest management, and environmental research**.

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3. Superpixel-Based Graph Representation

Satellite images often contain **millions of pixels**, making direct pixel-level analysis computationally expensive. To address this **challenge**, the image is first segmented into superpixels. A superpixel is a group of neighboring pixels that share similar characteristics such as color, texture, or intensity.

Instead of analyzing every pixel individually, each superpixel represents a **meaningful region** of the image. These regions are then treated as nodes in a graph structure, where connections between nodes represent spatial relationships between adjacent regions.

This representation **significantly reduces computational complexity** while preserving the important structural information in the image.

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4. Integration of Spectral and Spatial Features

Accurate satellite image analysis requires combining spectral information with spatial context.

- **Spectral features** come from the **reflectance values** of different bands (e.g., NDVI and other vegetation indices).
- **Spatial features** describe the relationship between neighboring regions, including shape, texture, and arrangement.

When these two types of information are integrated, the model gains a more comprehensive understanding of the scene. For example, two areas may have similar spectral values, but their **spatial arrangement and surrounding** context may reveal that one is a crop field while the other is a forest.

This integration improves classification accuracy and enhances the ability to interpret satellite imagery.

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5. Scalable Approach for Large Satellite Datasets

Satellite missions generate extremely large datasets covering vast geographic areas and long time periods. Processing such data efficiently requires scalable computational techniques.

The proposed approach improves scalability through:

- **Superpixel segmentation**, which reduces the number of data points.
- **Graph-based learning**, which captures relationships efficiently.
- **Batch processing and parallel computation**, enabling large-scale data analysis.

This allows the system to process large collections of satellite images from platforms such as **Earth observation missions**, making it suitable for real-world applications like agricultural monitoring, environmental assessment, and climate research.

REFERENCES

- *Xu, H., Zhang, S., & Wu, C. (2025).* ***Revealing the Impact of Urban Land Use Patterns on Land Surface Temperature Through Graph Attention Networks****. Sustainable Cities and Society.*

- *Lin, D., Lin, J., Zhao, L., Wang, Z. J., & Chen, Z. (2022).* ***Multilabel Aerial Image Classification With a Concept Attention Graph Neural Network****. IEEE Transactions on Geoscience and Remote Sensing.*

- *Dong, Y., Liu, Q., Du, B., & Zhang, L. (2022).* ***Weighted Feature Fusion of Convolutional Neural Network and Graph Attention Network for Hyperspectral Image Classification****. IEEE Transactions on Image Processing.*

- *Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2018).* ***Graph attention networks****. International Conference on* ***Learning Representations*** *(ICLR).*